Episode C3 | *Externalities*

The invisible hand can make mistakes.

This video introduces externalities and market failure.

Externalities

Markets do many things very well. We've looked at markets for goods where the seller carries all the costs of production and the buyer carries all the benefits of consumption. This is the generic good we've looked at so far.

We found markets do well with private goods. I can keep you from using mine.

Until this point we've talked about private goods goods that are rival. You using them as bad for me and excludable I keep.

An externality rises, when a person engages in an activity that influences the well-being of a bystander, but doesn't pay or receive compensation for their effort.

If I burn gasoline or dump industrial waste in a river, I'm negatively influencing the well-being of bystanders without compensating those who are affected.

If I get vaccinated or plant trees in my front yard I'm positively influencing the well-being of bystanders again without being compensated.

Lets go back to our friend, Molly, the small scale farmer. It turns out Molly uses a nitrogen fertilizer to make her land more productive.

The fertilizer isn't toxic but it does interfere in the waterways downstream when it runs off after a big rain.

Show Molly's farm

This is an example of a negative externality, where the cost associated with buying or selling a good aren't borne by those buying and selling the good.

Definition of externality

So ... thats the question ... Do markets opperate well when the costs aren't fully internalized?

Write the question on screen

Let's first start by ignoring Molly's externality, looking at the market.

Set up the supply and demand curves

In the past we've called the willingness to pay the Demand curve, but we can equivalently call it the marginal private benefit curve, since it represents the benefit for the buyer.

Label MPB

Similarly, the supply curve represents the willingness to supply gasoline at a range of prices, which we'll call the marginal private costs.

Label MPC

Without a considering the externality any quantity has a private benefit and a private cost. We represent these with the marginal private benefit and cost curves.

Zoom in on one quantity

Lets think back to our trusty immaginary social planner, who's concerned with maximizing the welfare of society.

We have a thought experiment tool we often use, called the social planner. This is a fake person who knows everyone's preferences and costs and tries to make the world as well off as possible. Of course this isn't ever going to be possible, but thinking from the perspective of the social planner often lets us think more clearly about puzzles like this.

So. How would the social planner choose the quantity when there are externalities?

I'd love an avatar of the social planner...

From this perspective, we can't ignore the costs created by the market borne by those outside the market. Both matter.

So from the social planner's perspective, we need to add the cost of Molly's nitrogent runoff to the model.

Add an externality on top of the cost line

This is what we call the marginal social cost.

Add the label to the vertical line

And in our little scenario here, there aren't externalized benefits. So the marginal social benefit is equal to the marginal private benefit.

Add an positive externality of zero, with a label on the other side of the vertical line

Since this externalized cost is associated with each item she's growing, we need to consider all the quantities she's sellling.

Sweep the line out to create the MPB, MPC, and MSC curves

And since we don't have externalized benefits in this scenario, the marginal social benefit curve is equal to the marginal private benefit curve.

Wiggle the MSB curve

So once we've fully considered the costs associated with the market by including the externalized costs to the cost curve, it's now easier to see that the marginal social cost will be higher than the marginal private costs.

Wiggle each one

Now that we've included costs appropriately, the social planner asks, how should I set quantity?

Question bubble

When there were no externalities the social planner maximizes welfare by choosing a quantity where the marginal cost and marginal benefit of the last unit are equal.

Highlight the MPC and MPB and show this point on the graph

But when we have externalized costs, the social planner wants to find the quantity that makes the marginal SOCIAL cost of the last unit equal to the marginal SOCIAL benefit.

Highlight the MSC and MSB

As always, we find the quantity by intersecting the two relevant curves.

Show the point

The difference here is that since we're including the externality. So we need to use the marginal social cost and benefits.

Wiggle the MSC and MSB

Intersecting the marginal social cost curve with a marginal social benefit curve we have no positive externalities we find the quantity that maximizes welfare with externalities.

Wiggle the equilibrium

And here we can see that this optimal quantity is smaller than the market equilibrium quantity.

Add a bracket showing the difference

This might fit with your intuition with comparative statics: quantity goes down when costs go up.

Welfare

Ok, but how does the social planner's choice influence welfare?

Speech bubble over social planner: I improved welfare, right?

Since we haven't said anything about prices, all we know is that consumer and producer surplus TOGETHER is the area between the marginal cost and marginal benefit curves. We'll just call this welfare and pin down prices later.

Point to prices

Welfare with externalities works very similarly to what we've done before. Except we're now looking at the costs and benefits of everyone, not just those buying and selling.

Zoom in on one quantity

The cost is under the marginal social cost curve, the benefit is below the marginal social benefit curve, and the welfare is the differece.

Highlight the cost, then the benefit, then the difference

And the marginal social cost is composed of both the marginal private cost, which is how costly the good is to produce, and the externality.

Highlight on the cost line the marginal private cost, then the externality

This is the socially efficient quantity. If we more quantity, we would increase costs more than we increase benefits.

Show this with a vertical line at some higher quantity

If we decrease by some quantity, we subtract benefits more than we subtract costs.

Show this with a vertical line at some lower quantity

Compare this to welfare in equilibrium while considering externalities.

Fade out welfare

And for dead weight loss think about adding one more unit of quantity.

We can find the marginal social benefit by intersecting the quantity or interested in with the marginal social benefit curve.

The same with marginal social costs.

So if the marginal social benefit is greater than the marginal social costs the unit is benefit.

So if the marginal social benefit is greater than the marginal social cost for a unit.

Adding that unit increases our welfare.

And if the marginal social benefit is less than the marginal social cost adding.

Adding that unit decreases social welfare.

For anyone unit, the unit carries a day weight loss if the benefit is greater than the cost and not exchanged.

Similarly.

Or the cost is greater than the benefit and exchanged. So dead weight loss is the area represented either by this missing benefit or by the added burden.

In all our previous economics, we looked only a deadly loss from lost surplus where the benefit was greater than the cost but not exchanged.

Here we have the other side where cost is greater than benefit, but we exchange it anyway.

So here we find the dead weight loss in our model by looking at where the marginal social cost is greater than the marginal social benefits above the difference between the social equilibrium quantity and the private equilibrium quantity.

This difference is what we call suboptimal quantity.

In the past we've called this lost quantity since we've only looked at deadway loss due to quantity exchanged being less than the efficient quantity.

Here we're extending that previous notion to include cases where quantity is too large.

A positive externality can be thought of in a similar way.

Flu shots or after it's developed a coveted vaccine carries a positive externality because I've a flu shot you the by-stander are less likely to get the flu.

So the marginal social benefit is greater than the marginal private benefit.

And here the social equilibrium quantity is higher than the private equilibrium quantity.

And one thing to note here is that even though we have externalities, the best thing for society once we've accounted for all the costs and benefits, the welfare maximizing quantity isn't necessarily zero!

If this type of modeling reminds you of taxes, you're on the right track.

A Corrective Tax

In the next video, we'll think about how to set prices properly and markets with externalities, which involves taxes and subsidies.